



FSF3827 Topics in Optimization, 2025
Assignment 4
Due Wednesday December 22 2025 23.59

Instructions for the assignments

- **Collaborations.** It is fine for students to collaborate on solving the assignment. However, every student needs to submit a report and if you have collaborated with someone you need to mention this clearly in the report. If you collaborate, it is essential that you all solve each task and fully understands all parts.
 - **Using AI / building upon code you found online.** It is fine to use an AI tool or build upon code that you find online. However, you need to fully understand the code and be able to discuss all parts of the code without help from AI. The teacher will ask questions and make sure you understand what the code does. If you use AI, I recommend you ask it to help with writing parts of the code or subroutines and not full programs.
 - The report should have a leading title page where the name, personal number, and e-mail address is clearly stated.
 - You are encouraged to write the reports in Latex.
 - Content
 1. The solutions must be described in enough details that other students (and the teacher) can easily read the report. For example, clearly define variables and sets used in the formulations.
 2. Describe the main parts of the problem formulations. For example, what does the constraints do.
 3. Describe the main parts of the code (what it does and how it works).
 - Everyone needs to upload on the Canvas page of the course no later than by the deadline of the assignment:
 - The report as a pdf file.
 - all the relevant code in a zip file.
 - You will be graded on a pass/fail scale. If the reports is insufficient, you will get second chance to revise the report.
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In this assignment you will work with some classical result from disjunctive programming and how to get a strong continuous relaxation.

Tasks

1. Consider the disjunctive constraint

$$\left[\left\| \begin{array}{cc} x_1 & 1 \\ x_2 & -1 \\ x_3 & 1 \end{array} \right\|_1 \leq 1 \right] \vee \left[\left\| \begin{array}{cc} x_1 & 9 \\ x_2 & -9 \\ x_3 & 9 \end{array} \right\|_1 \leq 1 \right] \vee \left[\left\| \begin{array}{cc} x_1 & 6 \\ x_2 & -1 \\ x_3 & 2 \end{array} \right\|_1 \leq 1 \right]$$

together with the variable bounds $0 \leq x_1, x_2, x_3 \leq 10$. First, write the disjunctive constraints without the 1-norm (only use linear constraints, and maybe some extra continuous variables).

- Draw the feasible set in the x_1, x_2 space (should be trivial).
- Form the big-M formulation of the disjunctive constraint, use the smallest valid M-values. Relax the binary variables as continuous variables between 0 and 1, and plot the feasible set in the x_1, x_2 space.
 - There are at least two ways you can do this. 1) Project out all the other variables using Fourier-Motzkin elimination. 2) Create a fine grid in the x_1, x_2 space, and check for each grid point if it can be feasible or not for the continuous relaxation (if it can be feasible then that grid point belong to the projected feasible set). In the second approach you fix x_1, x_2 to different values and solve a lot of optimization problems to check if the grid points are feasible.
- The big-M formulation typically gives a very weak continuous relaxation, but we can improve it. One approach is to lift in the other binary variables into each big-M constraint. You can apply the lifting procedure we used for cover cuts, but here there is a bit more structure.

In the big-M formulation you will have introduced three binary variables, let's call these y_1, y_2, y_3 , and you have the constraint $y_1 + y_2 + y_3 = 1$. Let's consider the following constraint from the big-M formulation

$$a^\top x \leq b + M(1 - y_1).$$

We can easily “lift in” y_2 and y_3 , if we first compute

$$\begin{aligned} M_1 &= \max a^\top x - b \\ \text{s.t. } & y_1 = 1 \\ & \text{and all other constraints in the big-M formulation,} \end{aligned}$$

$$\begin{aligned} M_2 &= \max a^\top x - b \\ \text{s.t. } & y_2 = 1 \\ & \text{and all other constraints in the big-M formulation,} \end{aligned}$$

$$\begin{aligned} M_3 &= \max a^\top x - b \\ \text{s.t. } & y_3 = 1 \\ & \text{and all other constraints in the big-M formulation.} \end{aligned}$$

A valid inequality is then given by

$$a^\top x \leq b + M_1 y_1 + M_2 y_2 + M_3 y_3$$

Apply this technique to all constraints in the big-M formulation (will result in the so-called improved big-M presented in

<https://doi.org/10.1016/j.compchemeng.2015.02.013>).

- Motivate why the resulting formulation can be stronger.
- Relax the binary variables as continuous variables between 0 and 1, and plot the feasible set in the x_1, x_2 space.
- For a big-M constraint with variable y_i in the original big-M formulation, do you need to compute the coefficient M_i or do you know what it will be?
- Is the resulting inequalities equivalent to what you obtain by the lifting procedure presented in Section 7.2 in "Integer Programming by Michele Conforti, Gérard Cornuéjols, Giacomo Zambelli (<https://link.springer.com/book/10.1007/978-3-319-11008-0>)? You can download the book through KTH (KTH network or VPN). Let me know if you can't access the book.

2. Read Chapter 4.9 in "Integer Programming by Michele Conforti, Gérard Cornuéjols, Giacomo Zambelli. (<https://link.springer.com/book/10.1007/978-3-319-11008-0>)

- Summarize the proof of Theorem 4.39.
- Write down a system of constraints that represents the convex hull of the disjunctive constraints in Task 1.

3. Read Chapter 3 in "Valid inequalities for mixed integer linear programs" by Gérard Cornuéjols. Read Chapter 3.3 extra carefully. (The paper is available on Canvas)

- Generate 6 rounds of lift-and-project cuts to the capacitated facility location problem described on the next pages. By 6 rounds, I mean solve LP relaxation, solve cut generating LP to derive a cut, add the cut to the problem and repeat 6 times. You can stop earlier if the LP relaxation gives an integer solution. How much does the relaxation improve?
- Challenge (you don't need to do this). Use the strengthened lift-and-project cuts from Chapter 3.4.

Good luck!

The text describing this class of problems has largely been generated by ChatGPT, and your teacher has made some modifications.

Problem Description

In the capacitated facility location (CFL) problem, we choose which facilities to open and how to ship product from opened facilities to customers. Each facility has a fixed opening cost and a limited capacity. Each customer has a fixed demand that must be fully satisfied. Shipping from a facility to a customer incurs a per-unit transportation cost. The goal is to minimize the total opening plus shipping cost.

Sets

- $I = \{1, 2, 3, 4\}$: set of customers
- $J = \{1, 2, 3, 4, 5\}$: set of candidate facilities

Parameters

- $d_i \geq 0$: demand of customer $i \in I$
- $u_j \geq 0$: capacity of facility $j \in J$
- $F_j \geq 0$: fixed opening cost of facility $j \in J$
- $c_{ij} \geq 0$: unit shipping cost from facility j to customer i

We assume the instance is feasible: $\sum_{j \in J} u_j \geq \sum_{i \in I} d_i$.

Variables

- $y_j \in \{0, 1\}$: equals 1 if facility j is opened, and 0 otherwise
- $x_{ij} \geq 0$: amount shipped from facility j to customer i

Mathematical Formulation

Objective

$$\min \sum_{j \in J} F_j y_j + \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \quad (1)$$

Constraints

Demand satisfaction (each customer fully served)

$$\sum_{j \in J} x_{ij} = d_i \quad \forall i \in I \quad (2)$$

Facility capacity (only if opened)

$$\sum_{i \in I} x_{ij} \leq u_j y_j \quad \forall j \in J \quad (3)$$

Variable domains

$$x_{ij} \geq 0 \quad \forall i \in I, \forall j \in J \quad (4)$$

$$y_j \in \{0, 1\} \quad \forall j \in J \quad (5)$$

Problem-Specific Data**Demands**

Customer	Demand d_i
1	15
2	10
3	12
4	13

Facility Capacities and Opening Costs

Facility	Capacity u_j	Opening cost F_j
1	20	140
2	15	110
3	25	160
4	19	120
4	21	110

Unit Shipping Costs

The unit shipping costs c_{ij} (customer i rows, facility j columns) are:

c_{ij}	Facility 1	Facility 2	Facility 3	Facility 4	Facility 5
Customer 1	3	6	5	8	6
Customer 2	7	4	6	5	5
Customer 3	5	7	4	2	6
Customer 4	6	5	7	4	1

Good luck!